Consistency and asymptotic normality of Maximum Likelihood Estimator (independent but not necessarily identically distributed)

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Newey and McFadden (1994) only includes theorems for MLE with i.i.d. assumption. I try to extend them with independent but not necessarily identically distributed assumption. The same asymptotic distribution is derived by Hoadley (1971), but not from Extremum estimator approach.

1 Consistency

Theorem 1 (Consistency of Extremum Estimator with compact parameter space)

Let $Q_0(\boldsymbol{\theta})$ be the limiting objective function and $\widehat{\boldsymbol{\theta}} := argmax_{\boldsymbol{\theta}\in\Theta}\widehat{Q}_n(\boldsymbol{\theta})$ i.e., the maximizer of the sample objective function. If (i) $\boldsymbol{\theta}_0$ is the unique maximizer of $Q_0(\boldsymbol{\theta})$. (ii) $\boldsymbol{\theta}_0 \in \Theta$ which is compact. (iii) $Q_0(\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$. (iv) $sup_{\boldsymbol{\theta}\in\Theta}||\widehat{Q}_n(\boldsymbol{\theta}) - Q_0(\boldsymbol{\theta})|| \to_p 0$. Then, $\widehat{\boldsymbol{\theta}} \to_p \boldsymbol{\theta}_0$.

Proof: See Theorem 2.1 of Newey and McFadden (1994). Q.E.D.

Lemma 2 (Information Inequality; Identification Conditions for MLE)

Let $f_i(y|\boldsymbol{x};\boldsymbol{\theta})$ be the conditional density and $\widehat{Q}_n(\boldsymbol{\theta}) = n^{-1}\sum_{i=1}^n lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$. If (i) $f_i(y|\boldsymbol{x};\boldsymbol{\theta}) \neq f_i(y|\boldsymbol{x};\boldsymbol{\theta}_0)$ for $\forall\boldsymbol{\theta} \in \Theta$ and $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$. (ii) $\mathbb{E}[|lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})|] < \infty$ for $\forall i$. (iii) $n^{-1}\sum_{i=1}^n \mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] := n^{-1}\sum_{i=1}^n Q_{0,i}(\boldsymbol{\theta}) \to Q_0(\boldsymbol{\theta})$. Then, $\boldsymbol{\theta}_0$ is the unique maximizer of $Q_{0,i}(\boldsymbol{\theta})$ and thus $Q_0(\boldsymbol{\theta})$.

Proof: Replace f with f_i in the Lemma 2.2 of Newey and McFadden (1994). Q.E.D.

Theorem 3 (Uniform LLN for independent but not necessarily identically distributed case)

If (i) $y_i|\boldsymbol{x}_i$ is independent but not necessarily identically distributed. (ii) parameter space Θ is compact. (iii) $a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$ for all $y_i|\boldsymbol{x}_i$ with probability one. (iv) For at least one $\delta > 0$, there exists $d_i(y_i|\boldsymbol{x}_i)$ such that $|a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})|^{1+\delta} \leq d_i(y_i|\boldsymbol{x}_i)$ for $\forall\boldsymbol{\theta} \in \Theta$ and $\mathbb{E}[d_i(y_i|\boldsymbol{x}_i)] < \infty$. Then, $sup_{\boldsymbol{\theta}\in\Theta}|n^{-1}\sum_{i=1}^n a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) - n^{-1}\sum_{i=1}^n \mathbb{E}[a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})]| \to_p 0$ and $n^{-1}\sum_{i=1}^n \mathbb{E}[a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})]$ is a continuous function of $\boldsymbol{\theta}$.

Reference: Theorem 4.2.2 of Amemiya (1985).

Theorem 4 (Consistency of MLE with compact parameter space)

If (i) $y_i|\boldsymbol{x}_i$ is independent but not necessarily identically distributed with conditional density $f_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)$. (ii) $f_i(y|\boldsymbol{x};\boldsymbol{\theta}) \neq f_i(y|\boldsymbol{x};\boldsymbol{\theta}_0)$ for $\forall\boldsymbol{\theta} \in \Theta$ and $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$. (iii) $\boldsymbol{\theta}_0 \in \Theta$ which is compact. (iv) $f_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$ for all

$y_i|\boldsymbol{x}_i$ with probability one. (v) $\mathbb{E}[sup_{\boldsymbol{\theta}\in\Theta}|lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})|^{1+\delta}] < \infty$ for at least one $\delta > 0$. (vi) $n^{-1}\sum_{i=1}^n \mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] := n^{-1}\sum_{i=1}^n Q_{0,i}(\boldsymbol{\theta}) \to Q_0(\boldsymbol{\theta})$. Then, $\widehat{\boldsymbol{\theta}} \to_p \boldsymbol{\theta}_0$.

Proof: (ii), (v) and (vi) implies Theorem 1's (i) by using Lemma 2 because (v) implies Lemma 2's (ii). (iii) is Theorem 1's (ii). Set $a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ and $d_i(y_i|\boldsymbol{x}_i) = sup_{\boldsymbol{\theta}\in\Theta}|a_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})|^{1+\delta}$ in Theorem 3. (i), (iii), (iv), and (v) is or implies Theorem 3's (i), (ii), (iii), and (iv) respectively. Thus, Theorem 3 implies Theorem 1's (iv) and (iii) since $n^{-1}\sum_{i=1}^n \mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] \to Q_0(\boldsymbol{\theta})$ by (vi). Q.E.D.

2 Asymptotic Normality

Theorem 5 (Asymptotic Normality of Extremum Estimator)

Let $\widehat{\boldsymbol{\theta}} := argmax_{\boldsymbol{\theta}\in\Theta}\widehat{Q}_n(\boldsymbol{\theta}) \to_p \boldsymbol{\theta}_0$. (i) $\boldsymbol{\theta}_0$ is not at the boundary of Θ. (ii) $\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta})$ exists and continuous in N where $\boldsymbol{\theta}_0 \in N$. (iii) $\sqrt{n}\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta}_0) \to_d N(\boldsymbol{0}, \boldsymbol{\Sigma})$. (iv) there exists $\boldsymbol{H}(\boldsymbol{\theta})$ such that it is continuous at $\boldsymbol{\theta}_0$ and $sup_{\boldsymbol{\theta}\in N}||\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta}) - \boldsymbol{H}(\boldsymbol{\theta})|| \to_p 0$. (v) $\boldsymbol{H}(\boldsymbol{\theta}_0)^{-1}$ exists. Then, $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \to_d N(\boldsymbol{0}, \boldsymbol{H}^{-1}\boldsymbol{\Sigma}\boldsymbol{H}^{-1})$.

Proof: See Theorem 3.1 of Newey and McFadden (1994). Q.E.D.

Theorem 6 (Liapounov Central Limit Theorem)

Let y_i be independent. If $lim_{n\to\infty}[\sum_{i=1}^n Var(y_i)]^{-1/2}(\sum_{i=1}^n \mathbb{E}[|y_i - \mathbb{E}[y_i]|^3])^{1/3} = 0$. Then, $\sqrt{n}(n^{-1}\sum_{i=1}^n y_i - n^{-1}\sum_{i=1}^n \mathbb{E}[y_i]) \to_d N(0, n^{-1}\sum_{i=1}^n Var(y_i))$.

Reference: Theorem 3.3.5 of Amemiya (1985).

Theorem 7 (Asymptotic Normality of MLE with compact parameter space)

If (i) $y_i|\boldsymbol{x}_i$ is independent but not necessarily identically distributed with conditional density $f_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)$ and other assumptions in Theorem 4 holds. (ii) $\boldsymbol{\theta}_0$ is not at the boundary of Θ. (iii) $f_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) > 0$ is twice continuously differentiable in N neighborhood of $\boldsymbol{\theta}_0$. (iv) Sufficient conditions for interchanging integration and differentiation. (v) $\boldsymbol{J}_i = \mathbb{E}[\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)']$ exists and is invertible. (vi) $\mathbb{E}[sup_{\boldsymbol{\theta}\in N}||\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})||^{1+\delta}] < \infty$ for at least one $\delta > 0$. (vii) $lim_{n\to\infty}[\sum_{i=1}^n \boldsymbol{J}_i]^{-1/2}(\sum_{i=1}^n \mathbb{E}[||\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)||^3])^{1/3} = 0$. Then, $\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \to_d N(\boldsymbol{0}, (n^{-1}\sum_{i=1}^n \boldsymbol{J}_i)^{-1})$.

Proof: (ii) is Theorem 5's (i). $\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}n^{-1}\sum_{i=1}^n lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) = n^{-1}\sum_{i=1}^n \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ and thus (iii) implies Theorem 5's (ii). $\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta}_0) = \nabla_{\boldsymbol{\theta}}n^{-1}\sum_{i=1}^n lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0) = n^{-1}\sum_{i=1}^n \nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)$. $\mathbb{E}[\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)] = \nabla_{\boldsymbol{\theta}}\mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)] = \nabla_{\boldsymbol{\theta}}Q_{0,i}(\boldsymbol{\theta}_0)$ by (iv). $\nabla_{\boldsymbol{\theta}}Q_{0,i}(\boldsymbol{\theta}_0) = 0$ because Theorem 4's (ii) and (v) implies $\boldsymbol{\theta}_0$ is the unique maximizer of $\mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] = Q_{0,i}(\boldsymbol{\theta})$. Thus, $Var(\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)) = \boldsymbol{J}_i$ by (v). With (i) and (vii), we apply Liapounov Central Limit Theorem, and have $\sqrt{n}\nabla_{\boldsymbol{\theta}}\widehat{Q}_n(\boldsymbol{\theta}_0) = \sqrt{n}(n^{-1}\sum_{i=1}^n \nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0) - 0) \to_d N(\boldsymbol{0}, n^{-1}\sum_{i=1}^n \boldsymbol{J}_i)$. Set $a_i(.) = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})$ and $d_i(.) = sup_{\boldsymbol{\theta}\in N}||\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})||^{1+\delta}$in Theorem 3. Theorem 4's (i), (iii), and this theorem's (iii) and (vi) implies $sup_{\boldsymbol{\theta}\in N}||n^{-1}\sum_{i=1}^n \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}) - n^{-1}\sum_{i=1}^n \mathbb{E}[\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})]|| \to_p 0$ and $n^{-1}\sum_{i=1}^n \boldsymbol{H}_i(\boldsymbol{\theta}) := n^{-1}\sum_{i=1}^n \mathbb{E}[\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}n^{-1}\sum_{i=1}^n \mathbb{E}[lnf_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta})] = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}n^{-1}\sum_{i=1}^n Q_{0,i}(\boldsymbol{\theta}) \to \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}Q_0(\boldsymbol{\theta}) := \boldsymbol{H}(\boldsymbol{\theta})$ is a continuous function of $\boldsymbol{\theta}$ and thus $\boldsymbol{H}(\boldsymbol{\theta})$ is continuous at $\boldsymbol{\theta}_0$. (iv) implies Information Matrix equality i.e., $\boldsymbol{J}_i = -\boldsymbol{H}_i$. So, $\boldsymbol{J} := n^{-1}\sum_{i=1}^n \boldsymbol{J}_i = -n^{-1}\sum_{i=1}^n \boldsymbol{H}_i := -\boldsymbol{H}$. Therefore, (v) implies Theorem 5's (v). Finally, $\boldsymbol{H}^{-1}\boldsymbol{J}\boldsymbol{H}^{-1} = \boldsymbol{J}^{-1}\boldsymbol{J}\boldsymbol{J}^{-1} = \boldsymbol{J}^{-1} = (n^{-1}\sum_{i=1}^n \boldsymbol{J}_i)^{-1}$. Q.E.D.

Remark: It is easy to check that the conditions of Theorem 7 implies the conditions of Theorem 2 in Hoadley (1971) for the asymptotic normality. For instance, N5 in Hoadley (1971) is $\mathbb{E}[\nabla_{\boldsymbol{\theta}} ln f_i(y_i|\boldsymbol{x}_i;\boldsymbol{\theta}_0)] = 0$ in our proof. N6 is Information Matrix equality, which is implied by (iv) of Theorem 7. N7 is $-n^{-1}\sum_{i=1}^n \boldsymbol{H}_i(\boldsymbol{\theta}) \to -\boldsymbol{H}(\boldsymbol{\theta})$ in our proof. N8 is Liapounov condition, which is Theorem 7's (vii). N9 is dominance condition, which is (vi) of Theorem 7. The asymptotic distribution is exactly the same. However, the conditions in Theorem 7 (I believe) is more primitive i.e., easier to interpret.

3 Application

Heterogeneous binary outcome model with $Pr(y_i = 1|\boldsymbol{x}_i) = F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$

$y_i|\boldsymbol{x}_i$ is independent and Bernoulli distributed with parameter $p_i = Pr(y_i = 1|\boldsymbol{x}_i) = F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$. Thus, it is not identically distributed since different i can have different $F_i(.)$ and thus p_i. By specifying conditions which implies the conditions in Theorem 4 and Theorem 7, $\widehat{\boldsymbol{\beta}}$ is consistent and asymptotic normal. In practice, $F_i(.)$ is usually chosen to be Logistic function or standard normal distribution function for $\forall i$. This model relaxes this assumption. From the latent variable approach of binary outcome model, $F_i(.)$ is the distribution function of the error of latent variable y_i^* e.g., utility level for making decision. It is natural to assume different individual has different distribution of random utility.

For example, we can extend Robit model (Liu, 2004) to heterogeneous Robit model $F_{t,\nu_i,\tau}(\boldsymbol{x}_i'\boldsymbol{\beta}_0) = F_{t,\nu_i}(\boldsymbol{x}_i'\boldsymbol{\beta}_0/\tau) = \int_{-\infty}^{\boldsymbol{x}_i'\boldsymbol{\beta}_0/\tau} t_{v_i}(z)dz$ where $t_{v_i}(.)$ is the density of standard t random variable with parameter ν_i. When $\tau = 1$ and $\nu_i \to \infty$, $t_{\nu_i}(.) \to \phi(.)$ and thus $F_{t,\nu_i,1}(.) \to \Phi(.)$ i.e., Probit model. When v_i is close to 7 and the scale parameter τ is 1.5484, $F_{t,7,1.5484}(.)$ is close to $Logistic(.)$ i.e., Logit model / Logistic regression. If ν_is are estimated simultaneously with $\boldsymbol{\beta}$, the data select the choice between Probit, Logit or mixture of them for each i within heterogeneous Robit model.

Theorem 8 (Consistency of MLE of heterogeneous binary outcome model)

If (i) $y_i|\boldsymbol{x}_i$ is independent but not necessarily identically distributed with conditional mass function $F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i}$. (ii) $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is positive definite (thus invertible). (iii) $\boldsymbol{\beta}_0 \in \Theta$ which is compact. (iv) $F_i(\boldsymbol{x}_i'\boldsymbol{\beta})$ is a continuous function of $\boldsymbol{\beta}$ with probability one and $F_i(.)$ is strictly monotone. (v) $\mathbb{E}[sup_{\boldsymbol{\beta}\in\Theta}|y_i lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1-y_i)ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}))|^{1+\delta}] < \infty$ for at least one $\delta > 0$. (vi) $n^{-1}\sum_{i=1}^n \mathbb{E}[y_i lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}) + (1-y_i)ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta})] \to Q_0(\boldsymbol{\beta})$. Then, $\widehat{\boldsymbol{\beta}} \to_p \boldsymbol{\beta}_0$.

Proof: Theorem 4's (ii) is implied by (ii) and (iv). $\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i']$ is positive definite. By the definition of positive definite, $(\boldsymbol{\beta}-\boldsymbol{\beta}_0)'\mathbb{E}[\boldsymbol{x}_i\boldsymbol{x}_i'](\boldsymbol{\beta}-\boldsymbol{\beta}_0) > 0$ for $\forall\boldsymbol{\beta} \neq \boldsymbol{\beta}_0 \implies \mathbb{E}[(\boldsymbol{x}_i'(\boldsymbol{\beta}-\boldsymbol{\beta}_0))^2] > 0$ for $\forall\boldsymbol{\beta} \neq \boldsymbol{\beta}_0 \implies \boldsymbol{x}_i'\boldsymbol{\beta} \neq \boldsymbol{x}_i'\boldsymbol{\beta}_0$ for $\forall\boldsymbol{\beta} \neq \boldsymbol{\beta}_0$. For strictly monotone $F_i(.)$, $F_i(\boldsymbol{x}_i'\boldsymbol{\beta}) \neq F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)$ for $\forall\boldsymbol{\beta} \neq \boldsymbol{\beta}_0 \implies F_i(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i} \neq F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^{y_i}(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^{1-y_i}$ for $\forall\boldsymbol{\beta} \neq \boldsymbol{\beta}_0$. Other conditions are direct application of Theorem 4. Q.E.D.

Remark: (v) can be replaced by $1 - F_i(z) = F_i(-z)$ and $|lnF_i(z)|^{1+\delta} < \infty$ for $\forall z$ and for at least one finite $\delta > 0$.

$$
\begin{aligned}
|y_i lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0) + (1-y_i)ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))|^{1+\delta} &= 2^{1+\delta}|\frac{1}{2}[y_i lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0) + (1-y_i)ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))]|^{1+\delta} \\
&\leq 2^{1+\delta}[\frac{1}{2}|y_i lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)|^{1+\delta} + \frac{1}{2}|(1-y_i)ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))|^{1+\delta}] \\
&\qquad\text{By Jensen inequality} \\
&= 2^{\delta}[|y_i|^{1+\delta}|lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)|^{1+\delta} + |1-y_i|^{1+\delta}|ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))|^{1+\delta}] \\
&\leq 2^{\delta}[1\cdot|lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)|^{1+\delta} + 1\cdot|ln(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))|^{1+\delta}] \\
&= 2^{\delta}[|lnF_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)|^{1+\delta} + |lnF_i(-\boldsymbol{x}_i'\boldsymbol{\beta}_0)|^{1+\delta}] < \infty
\end{aligned}
$$

Theorem 9 (Asymptotic normality of MLE of heterogeneous binary outcome model)

If (i) $y_i|\boldsymbol{x}_i$ is independent but not necessarily identically distributed and other assumptions in Theorem 8 holds. (ii) $\boldsymbol{\beta}_0$ is not at the boundary of Θ. (iii) $F_i(\boldsymbol{x}_i'\boldsymbol{\beta})$ is twice continuously differentiable in N neighborhood of $\boldsymbol{\beta}_0$. (iv) Sufficient conditions for interchanging integration and differentiation. (v) $\boldsymbol{J}_i = \mathbb{E}[\frac{1}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i']$ exists and is invertible. (vi) $\mathbb{E}[sup_{\boldsymbol{\beta}\in N}||\{\frac{y_i-F_i}{F_i(1-F_i)}F_i^{''} - \frac{(y_i-F_i)^2}{F_i^2(1-F_i)^2}F_i'^2\}\boldsymbol{x}_i\boldsymbol{x}_i'||^{1+\delta}] < \infty$ for at least one $\delta > 0$.
(vii) $lim_{n\to\infty}[\sum_{i=1}^n \boldsymbol{J}_i]^{-1/2}(\sum_{i=1}^n \mathbb{E}[||\frac{y_i-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i||^3])^{1/3} = 0$. Then, $\sqrt{n}(\widehat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) \to_d N(\boldsymbol{0}, (n^{-1}\sum_{i=1}^n \boldsymbol{J}_i)^{-1})$.

Proof: (iii) implies $F_i(\boldsymbol{x}_i'\boldsymbol{\beta})^{y_i}(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}))^{1-y_i}$ is twice continuously differentiable in N neighborhood of $\boldsymbol{\beta}_0$. The derivation of $\boldsymbol{J}_i$ is

$$\begin{aligned}
\boldsymbol{J}_i &= \mathbb{E}[\frac{y_i - F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i\{\frac{y_i - F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i\}'] \\
&= \mathbb{E}[\mathbb{E}[\frac{y_i - F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i\{\frac{y_i - F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)\boldsymbol{x}_i\}'|\boldsymbol{x}_i] \\
&= \mathbb{E}[\frac{\mathbb{E}[y_i - F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)|\boldsymbol{x}_i]}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))^2}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i'] \\
&= \mathbb{E}[\frac{1}{F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0)(1-F_i(\boldsymbol{x}_i'\boldsymbol{\beta}_0))}F_i'(\boldsymbol{x}_i'\boldsymbol{\beta}_0)^2\boldsymbol{x}_i\boldsymbol{x}_i']
\end{aligned}$$

which is positive semi-definite (Wooldridge, 2010, p.568). For (vi), define $F_i := F_i(\boldsymbol{x}_i'\boldsymbol{\beta})$,

$$\begin{aligned}
\nabla_{\boldsymbol{\beta}}\nabla_{\boldsymbol{\beta}}\{y_i lnF_i + (1-y_i)ln(1-F_i)\} &= -(\frac{y_i}{F_i^2} + \frac{1-y_i}{(1-F_i)^2})F_i'^2\boldsymbol{x}_i\boldsymbol{x}_i' + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})F_i^{''}\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= [-(\frac{y_i}{F_i^2} + \frac{1-y_i}{(1-F_i)^2})F_i'^2 + (\frac{y_i}{F_i} - \frac{1-y_i}{1-F_i})F_i^{''}]\boldsymbol{x}_i\boldsymbol{x}_i' \\
&= [-\frac{(y_i-F_i)^2}{F_i^2(1-F_i)^2}F_i'^2 + \frac{y_i-F_i}{F_i(1-F_i)}F_i^{''}]\boldsymbol{x}_i\boldsymbol{x}_i' \qquad \text{since } y_i = y_i^2
\end{aligned}$$

Q.E.D.

4 Reference

Amemiya, T. (1985). Advanced Econometrics, Harvard University Press.

Liu, C. (2004). Robit Regression: A Simple Robust Alternative to Logistic and Probit Regression. In Applied Bayesian Modeling and Causal Inference from Incomplete-Data Perspectives (eds W.A. Shewhart, S.S. Wilks, A. Gelman and X.-L. Meng).

Hoadley, B. (1971). Asymptotic Properties of Maximum Likelihood Estimators for the Independent Not Identically Distributed Case. The Annals of Mathematical Statistics, 42(6), 1977–1991.

Newey, W., & McFadden, D. (1994). Large Sample Estimation and Hypothesis Testing. Handbook of Econometrics, 4, 2111-2245.

Wooldridge, J. M. (2010). Econometric Analysis of Cross Section and Panel Data (2nd Edition).